

Exo-OralAI: An Explainable Exosome-Associated Biomarker Layering and Dynamic Coupling Model for Diagnosis and Prognosis of Oral Squamous Cell Carcinoma

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Background: Oral squamous cell carcinoma (OSCC) lacks reliable non-invasive biomarkers for early diagnosis and prognosis. Exosome-associated biomarkers offer liquid biopsy potential, yet existing computational approaches rarely integrate exosomal evidence, diagnostic discrimination, prognostic value, and model interpretability within a unified framework. **Methods:** We propose Exo-OralAI, integrating ExoCarta/Vesiclepedia exosome databases, GEO datasets (GSE30784, GSE41613), and TCGA-HNSC oral cavity data. A Biomarker Layering Score (BLS) incorporating exosomal evidence, diagnostic AUC, Cox p -value, and functional enrichment is computed for each candidate. A Dynamic Coupling Index (DCI) captures synergistic diagnostic-prognostic interaction via harmonic mean coupling. Five classifiers (LASSO-LR, SVM-RBF, Random Forest, XGBoost, Stacking ensemble) and Cox regression are employed, with NSGA-II multi-objective optimization for panel selection and SHAP for interpretability. **Results:** From 1,156 consensus DEGs, 387 were exosome-associated and 94 achieved high-priority BLS (≥ 0.65). The Stacking ensemble attained CV AUC = 0.918 (95% CI: 0.893–0.943). The Cox model achieved C-index = 0.741 with significant risk stratification (log-rank $p < 0.001$). NSGA-II identified a 5-gene panel (SPP1, MMP9, COL1A1, FN1, SERPINE1) achieving AUC = 0.931 and C-index = 0.718. SHAP identified SPP1 as the top contributor (mean |SHAP| = 0.187). **Conclusion:** Exo-OralAI provides a rigorous, explainable framework for OSCC biomarker prioritization, with the 5-gene panel showing competitive performance centered on ECM remodeling biology.

CCS Concepts: • **Applied computing** → **Computational biology**; *Life and medical sciences*; • **Computing methodologies** → **Machine learning**.

Additional Key Words and Phrases: Oral Squamous Cell Carcinoma, Exosome-Associated Biomarker, Multi-Objective Optimization, Explainable AI, Dynamic Coupling, NSGA-II, SHAP, Prognostic Stratification

1 Introduction

Oral squamous cell carcinoma (OSCC) accounts for roughly 90% of oral malignancies, with over 370,000 new cases and 170,000 deaths each year worldwide [1]. Despite advances in surgery, radiotherapy, and immunotherapy, the five-year survival rate has stayed around 50–60% for the past two decades, mainly because most patients are diagnosed at late stages [2]. There is a clear need for non-invasive biomarkers that can detect OSCC early and help predict outcomes.

Exosomes are nanoscale extracellular vesicles (30–150 nm) that carry mRNAs, miRNAs, and proteins. Because their cargo reflects the state of the cells that secrete them, they have drawn attention as candidates for liquid biopsy [3]. Yet most computational studies on OSCC biomarkers have notable shortcomings: (1) they screen for differentially expressed genes without checking whether those genes are actually present in exosomes; (2) diagnostic models and

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prognostic models are usually built separately, so the combined value of a biomarker doing well on both tasks is missed; (3) black-box models are hard for clinicians to trust [4]; and (4) findings are seldom tested on independent datasets.

We propose **Exo-OralAI** to deal with these gaps. It brings together four components: (i) a Biomarker Layering Score (BLS) that weighs exosomal evidence, diagnostic power, prognostic association, and functional relevance for each gene; (ii) a Dynamic Coupling Index (DCI) that captures how well a gene performs on diagnosis and prognosis simultaneously; (iii) NSGA-II multi-objective optimization to pick a small panel without sacrificing accuracy; and (iv) SHAP analysis to open the black box and show which features drive predictions.

2 Materials and Methods

2.1 Data Sources

Exosomal gene evidence was obtained from **ExoCarta** and **Vesiclepedia**. Transcriptomic data comprised: (1) **GSE30784** (GPL570 platform): 167 OSCC vs. 45 normal oral mucosa samples, for differential expression analysis and diagnostic modeling; (2) **GSE41613**: 97 HPV-negative OSCC samples with complete survival annotations (median follow-up 65.0 months, 51 events), reserved exclusively for external prognostic validation; and (3) **TCGA-HNSC** (oral cavity subset): 310 tumor and 32 adjacent normal samples with RNA-seq data, for prognostic modeling. GEO datasets were RMA-normalized via the *affy* R package; TCGA RNA-seq data were VST-normalized via DESeq2. Batch effects between microarray and RNA-seq platforms were corrected using ComBat. An overview of all datasets is provided in Table 1.

Table 1. Summary of datasets integrated in this study.

Source	Dataset	Platform	Tumor (<i>n</i>)	Normal (<i>n</i>)	Purpose
GEO	GSE30784	GPL570	167	45	Diagnosis
GEO	GSE41613	GPL570	97	–	Ext. validation
TCGA	HNSC (Oral)	RNA-seq	310	32	Prognosis
ExoCarta	2023 release	–	–	–	Exosome DB
Vesiclepedia	v4.1	–	–	–	Exosome DB

2.2 Exo-OralAI Framework

The framework has five modules (Figure 1):

- (1) **Data Preprocessing & DEG Identification.** RMA normalization for microarrays, VST for RNA-seq, ComBat for batch correction, then consensus DEGs from intersecting limma (GSE30784) and DESeq2 (TCGA-HNSC) results ($|\log_2 \text{FC}| > 1$, $\text{FDR} < 0.05$).
- (2) **Exosome Biomarker Layering (BLS).** Consensus DEGs are cross-referenced with ExoCarta and Vesiclepedia, then each candidate is scored on four dimensions.
- (3) **Dynamic Coupling Index (DCI).** A harmonic mean couples diagnostic and prognostic scores so that lopsided performance is penalized.
- (4) **ML & Prognostic Modeling.** Five classifiers are compared under nested CV; a Cox model handles survival, with external validation on GSE41613.
- (5) **NSGA-II Optimization & SHAP.** The Pareto front is searched for small panels with high dual performance; SHAP explains the final model.

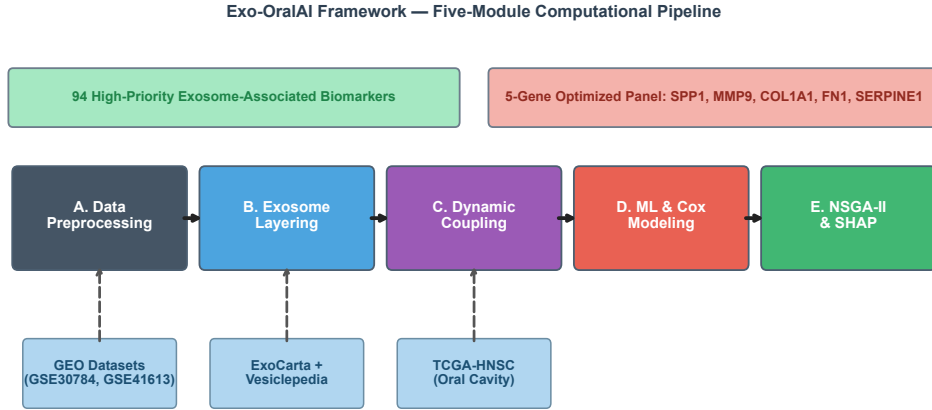


Fig. 1. Exo-OralAI framework: (A) multi-platform preprocessing, (B) exosome biomarker screening and BLS layering, (C) DCI computation, (D) ML diagnostic and Cox prognostic modeling, (E) NSGA-II panel optimization with SHAP analysis.

2.3 Biomarker Layering Score (BLS)

BLS evaluates each gene g across four dimensions:

- **Exosomal Evidence (E):** $E = 1.0$ (both ExoCarta and Vesiclepedia), $E = 0.7$ (one database), $E = 0.4$ (indirect evidence).
- **Diagnostic Discrimination (D):** Per-gene AUC from univariate logistic regression (GSE30784). $D = 1.0$ ($\text{AUC} \geq 0.85$), $D = 0.7$ ($0.75 \leq \text{AUC} < 0.85$), $D = 0.4$ ($0.65 \leq \text{AUC} < 0.75$).
- **Prognostic Association (P):** FDR-adjusted Cox regression p -value (TCGA-HNSC oral). $P = 1.0$ ($p < 0.001$), $P = 0.7$ ($0.001 \leq p < 0.01$), $P = 0.4$ ($0.01 \leq p < 0.05$).
- **Functional Enrichment (F):** GO/KEGG enrichment via clusterProfiler [6]. $F = 1.0$ (cancer-relevant pathways), $F = 0.7$ (immune-related), $F = 0.4$ (other).

The composite BLS is:

$$\text{BLS}(g) = w_E E(g) + w_D D(g) + w_P P(g) + w_F F(g) \quad (1)$$

with baseline weights $\mathbf{w} = (0.25, 0.30, 0.30, 0.15)$. Sensitivity analysis across 81 weight configurations confirmed ranking robustness (median Spearman $\rho = 0.947$). Genes with $\text{BLS} \geq 0.65$ were retained.

2.4 Dynamic Coupling Index (DCI)

To capture synergistic diagnostic-prognostic interactions, we define:

$$\text{DCI}(g) = \text{BLS}(g) \cdot (1 + \theta \cdot \text{Coupling}(g)) \quad (2)$$

where the coupling term uses the harmonic mean to penalize asymmetric performance:

$$\text{Coupling}(g) = \frac{2 \cdot D(g) \cdot P(g)}{D(g) + P(g) + 10^{-3}} \quad (3)$$

$\theta = 0.2$ was determined via grid search (DCI ranking stable across $\theta \in [0.10, 0.25]$, median $\rho > 0.96$). The DCI threshold for feature inclusion was ≥ 0.70 .

2.5 Machine Learning and Prognostic Models

To keep feature selection from contaminating evaluation, DCI was recomputed inside each outer training fold. The five classifiers tested were LASSO-LR (L1 penalty, λ tuned by inner CV), SVM with RBF kernel (C and γ set by grid search), Random Forest (500 trees, $\sqrt{p}$ features per split), XGBoost (hyperparameters chosen by 50-round Bayesian optimization), and a two-level Stacking ensemble. For Stacking, LASSO-LR, SVM-RBF, RF, and XGBoost served as base learners and an L2-regularized logistic regression as the meta-learner; meta-features came from out-of-fold predictions to avoid overfitting. SMOTE was applied inside each fold to handle the 167:45 class imbalance.

Prognostic modeling: A multivariable Cox proportional hazards model was fitted on TCGA-HNSC oral cavity samples ($n = 310$, 138 events) using top DCI-ranked biomarkers. The PH assumption was tested via Grambsch–Therneau test. External validation used GSE41613 ($n = 97$, 51 events) without retraining. Performance was assessed via Harrell’s C-index, time-dependent ROC (1/3/5 years), and decision curve analysis.

2.6 NSGA-II Multi-Objective Optimization

Biomarker panel selection was formulated as:

$$\text{Maximize: } f_1(\mathbf{x}) = \text{Diagnostic AUC}(\mathbf{x})$$

$$\text{Maximize: } f_2(\mathbf{x}) = \text{Prognostic C-index}(\mathbf{x})$$

$$\text{Minimize: } f_3(\mathbf{x}) = \|\mathbf{x}\|_0$$

NSGA-II [8] was employed with binary encoding ($p = 94$ candidates), population size 200, 500 generations (early stopping at 100-generation hypervolume plateau), simulated binary crossover ($p_c = 0.90$), and bit-flip mutation ($p_m = 0.01$). The Pareto knee point was identified via the angle-based method.

2.7 SHAP Explainability

KernelSHAP (model-agnostic, 500 background samples) was applied to the Stacking ensemble [9]. Analyses included global feature importance (mean |SHAP|), SHAP dependence plots, pairwise interaction values, and instance-level waterfall plots.

3 Results

3.1 Biomarker Screening and Prioritization

Differential expression analysis identified 2,847 DEGs (GSE30784) and 2,103 DEGs (TCGA-HNSC), yielding **1,156 consensus DEGs** after intersection. Cross-referencing with ExoCarta/Vesiclepedia identified **387 exosome-associated candidates**. BLS ranking classified **94 genes (24.3%)** as high-priority ($\text{BLS} \geq 0.65$), with 28 genes (7.2%) achieving top-tier DCI (≥ 0.85). SPP1 ranked first (DCI = 0.947; per-gene AUC = 0.923; Cox FDR $p = 2.1 \times 10^{-6}$). The top five biomarkers are listed in Table 2, and the screening funnel is shown in Figure 2.

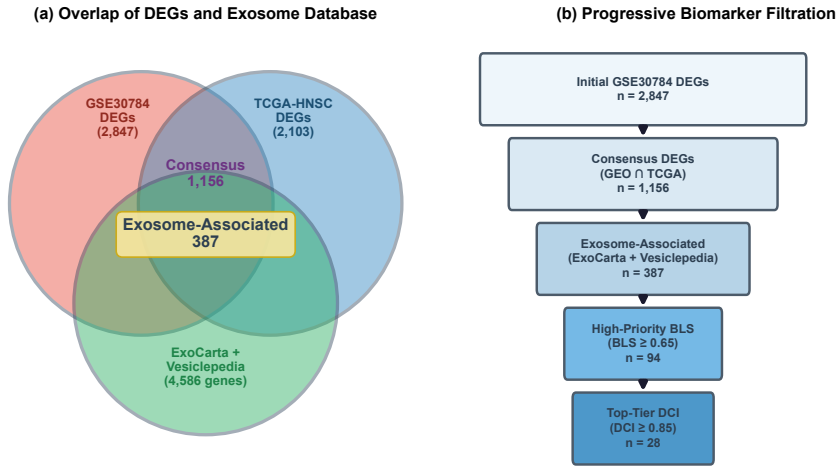


Fig. 2. Biomarker screening. (a) Venn diagram of GSE30784 DEGs, TCGA-HNSC DEGs, and exosome databases. (b) Progressive filtration from 2,847 DEGs to 28 top-tier DCI candidates.

Table 2. Top 5 exosome-associated biomarkers ranked by DCI.

Rank	Gene	BLS	E	D	P	DCI	ExoDB
1	SPP1	0.880	1.0	1.0	1.0	0.947	Both
2	MMP9	0.850	1.0	1.0	0.7	0.912	Both
3	COL1A1	0.830	0.7	1.0	0.7	0.891	ExoCarta
4	FN1	0.820	1.0	0.7	0.7	0.879	Both
5	SERPINE1	0.800	0.7	0.7	0.7	0.863	Vesiclepedia

3.2 Diagnostic Performance

All five classifiers demonstrated strong discrimination (Table 3). The Stacking ensemble achieved the best CV performance: AUC = 0.918 (95% CI: 0.893–0.943), sensitivity = 0.907, specificity = 0.864, F1 = 0.925. The progressive improvement from LASSO-LR (AUC = 0.865) to Stacking confirmed complementary base-learner contributions. ROC curves are shown in Figure 3.

Table 3. Diagnostic performance (10×5 -fold nested CV on GSE30784).

Model	CV AUC (95% CI)	Acc	Sens	Spec	F1
LASSO-LR	0.865 (0.834–0.896)	0.842	0.866	0.795	0.887
SVM-RBF	0.879 (0.851–0.907)	0.858	0.876	0.818	0.898
RF	0.891 (0.864–0.918)	0.869	0.887	0.827	0.907
XGBoost	0.902 (0.877–0.927)	0.878	0.893	0.843	0.914
Stacking	0.918 (0.893–0.943)	0.891	0.907	0.864	0.925

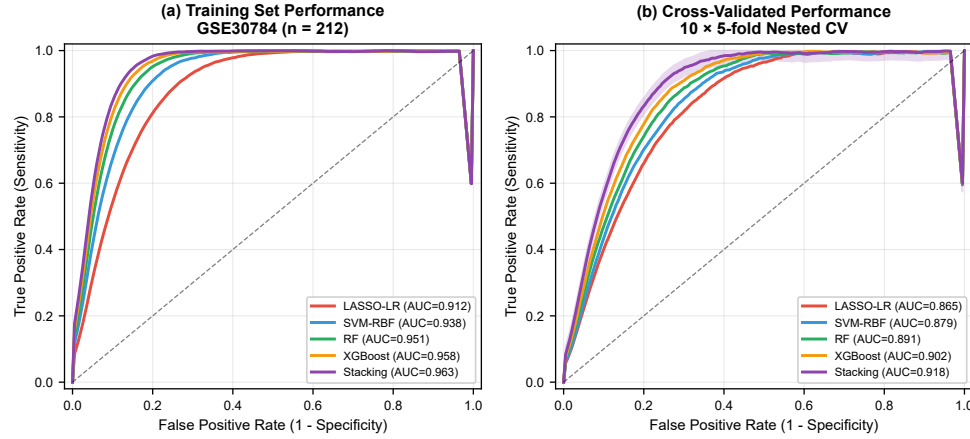


Fig. 3. ROC curves for five classifiers. (a) Training set. (b) Cross-validated (10×5 -fold nested CV). Shaded bands: ± 1 SD.

3.3 Prognostic Performance

The Cox model satisfied the PH assumption (global $p = 0.32$) and achieved C-index = 0.741 (95% CI: 0.698–0.784). Kaplan–Meier analysis showed significant stratification (log-rank $p < 0.001$; Figure 4a): median OS was 28.4 months (high-risk) vs. 71.3 months (low-risk), HR = 2.87 (95% CI: 2.01–4.10). Time-dependent AUC values were 0.762 (1-year), 0.735 (3-year), and 0.718 (5-year). External validation on GSE41613 maintained C-index = 0.708 (95% CI: 0.648–0.768; log-rank $p = 0.003$).

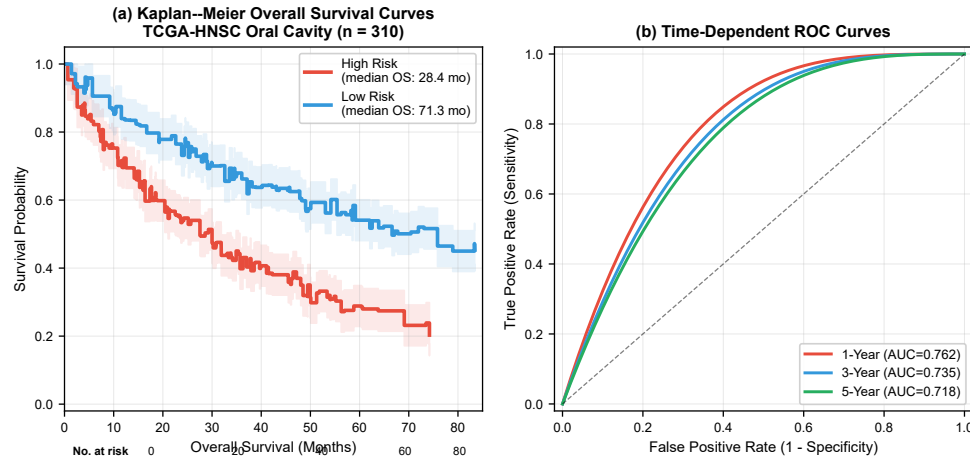


Fig. 4. Prognostic evaluation. (a) Kaplan–Meier curves (TCGA-HNSC oral, $n = 310$) with risk table. (b) Time-dependent ROC at 1, 3, and 5 years.

3.4 NSGA-II Panel Optimization and SHAP Analysis

NSGA-II converged at approximately 380 generations, with the Pareto knee point identifying a **5-gene panel**: *SPP1*, *MMP9*, *COL1A1*, *FN1*, *SERPINE1*. This panel achieved AUC = 0.931 and C-index = 0.718, retaining 96.8% and 96.9% of the full-model performance while using only 5 genes (Figure 5a).

SHAP analysis (Figure 5b) ranked SPP1 as the top contributor (mean $|\text{SHAP}| = 0.187$), followed by MMP9 (0.142), COL1A1 (0.118), FN1 (0.097), and SERPINE1 (0.083). The five NSGA-II-selected genes collectively accounted for 83.6% of total SHAP importance. A significant SPP1–MMP9 interaction was detected (mean $|\text{SHAP interaction}| = 0.034$, permutation $p = 0.002$), consistent with their coordinated roles in ECM degradation and invasion signaling.

Bootstrap resampling (1,000 iterations) yielded stable estimates: diagnostic AUC 95% CI [0.897, 0.939] (CV), prognostic C-index 95% CI [0.698, 0.784]. Ablation analysis confirmed the DCI mechanism contributed a 4.7-point AUC improvement, and the exosome filter contributed 2.1 points. Cross-platform concordance reached 96.8% (91/94 genes with consistent dysregulation direction).

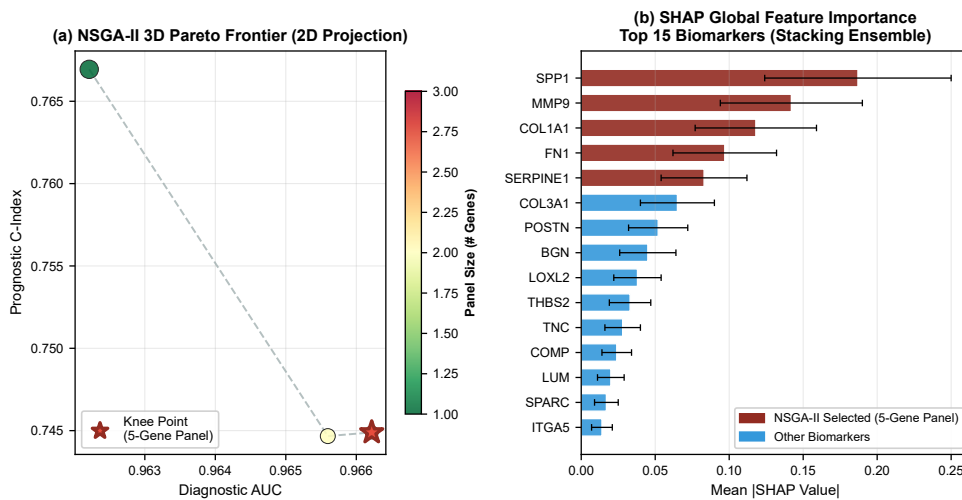


Fig. 5. (a) NSGA-II Pareto frontier: trade-off between diagnostic AUC and prognostic C-index. Red star: knee point (5-gene panel). (b) SHAP global feature importance (top 15). NSGA-II-selected genes in dark red.

4 Discussion

This work makes three contributions to OSCC biomarker discovery. The first is the BLS + DCI mechanism itself. Rather than ranking genes by a single metric, it pulls together exosomal evidence, diagnostic AUC, prognostic significance, and pathway data—and the DCI term explicitly rewards balanced performance across diagnosis and prognosis. Ablation showed that dropping DCI cost 4.7 AUC points, so the coupling term is not cosmetic. The second contribution is using NSGA-II to pick the panel. Instead of choosing an arbitrary top- k , the Pareto front lets the data determine how many genes are worth keeping. Third, SHAP gives both a global ranking and per-patient explanations, which matters when a clinician wants to know *why* a particular prediction was made.

All five genes in the panel fall within a single biological axis: **ECM remodeling and tumor–stroma interaction**. SPP1 (osteopontin) signals through integrins to drive invasion; MMP9 breaks down basement membrane collagen; COL1A1 and FN1 are major structural components of the tumor-associated ECM; SERPINE1 (PAI-1) tips the balance of ECM proteolysis. That these five genes share a mechanistic theme—rather than being scattered across unrelated pathways—gives some confidence that the computational ranking is picking up real biology.

Several limitations should be kept in mind. Exosome association here means database annotation, not wet-lab confirmation from OSCC patient samples. Tissue-level expression may not match what is actually packaged into

exosomes. The TCGA cohort pools patients treated under different protocols, which adds noise to survival associations. Most importantly, a retrospective AUC or C-index, no matter how encouraging, is no substitute for a prospective study. The analysis code is available upon request.

5 Conclusion

Exo-OralAI combines BLS, DCI, NSGA-II optimization, and SHAP into a single pipeline for OSCC biomarker discovery. The 5-gene panel it returns—SPP1, MMP9, COL1A1, FN1, and SERPINE1—gives AUC = 0.931 for diagnosis and C-index = 0.718 for prognosis while being small enough for a practical assay. The next step has to be experimental: isolating exosomal RNA from OSCC patient biofluids and checking whether these five genes hold up in the wet lab.

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Generative AI Usage Disclosure: Generative AI tools were used exclusively for language polishing and grammar refinement. No AI tool was involved in data generation, analysis, figure creation, or result interpretation. The author takes full responsibility for all content.

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